EV Technology Course — Module 5: Vehicle Control, Motor Control and Regenerative Braking

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An EV's driving feel is written in software. The vehicle control unit (VCU) turns pedal positions into torque requests, the motor controller turns torque requests into precise currents thousands of times a second, and the brake controller decides how much braking the motor can recover before the friction brakes join in. This module explains each layer.

1. Learning objectives

- Explain the control hierarchy: VCU, motor controller (MCU), BMS and brake/ESC controller.
- Explain torque arbitration, drive modes, creep and one-pedal driving.
- Explain field-oriented control (FOC) of a PMSM in plain terms.
- Explain regenerative braking and brake blending, and what limits regen.
- Explain traction control, stability control and torque vectoring in EVs.
- Calculate regen power, friction share and wheel slip.

2. Fundamentals

Think of three layers:

1. **Decide (VCU):** "What torque should the wheels get right now?" It weighs the driver's pedal, drive mode, battery limits, temperatures, traction and stability requests.
2. **Execute (MCU):** "What currents make exactly that torque?" It controls the inverter switches to produce the requested torque within milliseconds.
3. **Protect (BMS, ESC, thermal):** "What must not happen?" They set limits the VCU must respect.

All layers talk over CAN or Ethernet (Module 10).

3. Detailed technical explanation

VCU functions

Function	What it does
Pedal map	Converts pedal position and speed into a torque request; differs per drive mode (Eco, Normal, Sport)
Torque arbitration	Takes the minimum of driver request, BMS discharge/charge limits, motor/inverter thermal limits, ESC interventions
Creep	Small torque at standstill to mimic an automatic car
One-pedal / regen levels	Lift-off deceleration set by regen level (often up to ~0.2–0.3 g)
Energy management	Splits power between traction, HVAC and auxiliaries; range estimation
State machine	Sleep, wake, ready, drive, charge, fault modes; contactor sequencing with the BMS

Field-oriented control (FOC). The MCU measures phase currents and rotor angle, mathematically rotates the three AC currents into two DC-like values: i_d (along the magnet) and i_q (across the magnet, which makes torque). Two simple PI controllers hold i_d and i_q at their targets; the result is turned back into three PWM duty cycles. This makes an AC motor as easy to control as a DC motor, with fast, smooth torque response. Field weakening is done by commanding negative i_d.

Regenerative braking

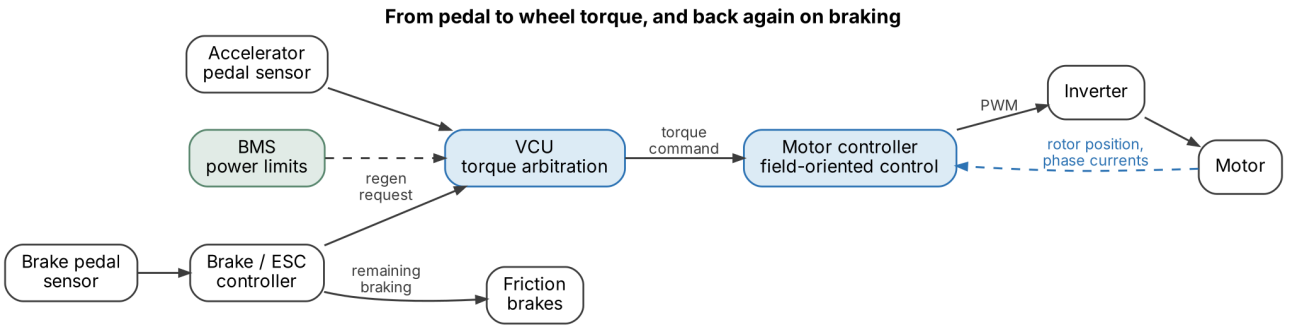
- When the driver lifts off or brakes, the VCU asks for negative torque; the motor acts as a generator.
- Regen is limited by: maximum motor/inverter torque and power, **battery charge acceptance** (low when full or cold), vehicle stability (rear-axle regen on slippery roads), and very low speed (regen fades below ~5–10 km/h).
- **Brake blending:** the brake controller splits total braking demand between regen and friction brakes so the driver feels a consistent pedal. This needs an electric brake booster or brake-by-wire (Module 9).
- Real-world regen typically recovers 15–30% of the energy used in city driving.

Traction, stability and torque vectoring

- EV motors change torque in a few milliseconds (versus 100+ ms for an engine), so traction control can be much finer.

- Wheel slip is detected from wheel-speed sensors; the VCU/MCU cuts or adds torque directly at the motor.
- Dual-motor cars split torque front/rear; cars with two rear motors can send different torque to left and right wheels (torque vectoring) for agility and stability.

4. Diagram: from pedal to wheel torque, and back on braking



5. Step-by-step: a city stop from 60 km/h

1. Driver lifts off the accelerator: the VCU commands regen torque according to the selected regen level.
2. The motor generates; current flows back through the inverter into the battery (if the BMS allows).
3. Driver presses the brake pedal: the brake controller computes total deceleration demanded.
4. It asks the VCU for as much regen as possible; the VCU returns how much it can deliver (limited by battery and motor).
5. Friction brakes supply the remainder; at very low speed regen fades and friction takes over smoothly.
6. If a wheel locks or slips, ABS/ESC reduce regen and modulate friction brakes on that wheel.
7. At standstill, a hold function or creep torque keeps the car stable.

6. Engineering equations

Braking force and regen power:

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F_b = m \times a \qquad P_{regen,wheel} = F_b \times v
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Power reaching the battery:

$$P_{bat} = P_{regen,wheel} \times \eta_{gear} \times \eta_{motor} \times \eta_{inv}$$

Wheel slip ratio (driving):

$$\lambda = \frac{\omega r - v}{v}$$

Park transform (FOC, for reference):

$$i_d = i_\alpha \cos\theta + i_\beta \sin\theta \qquad i_q = -i_\alpha \sin\theta + i_\beta \cos\theta$$

a in m/s² (0.1 g = 0.981 m/s²), v in m/s, ω = wheel angular speed (rad/s), θ = rotor electrical angle, i_α/i_β = two-axis stator currents from the Clarke transform.

7. Worked examples

Example 1: regen at 0.2 g from 80 km/h. m = 1,700 kg. F = 1,700 × 9.81 × 0.2 = **3,335 N**; at 22.2 m/s, P = **74 kW** at the wheels. If the system can take only 60 kW, friction brakes provide the other **14 kW**.

Example 2: wheel slip. Driven wheel surface speed 25 m/s, vehicle 22 m/s: λ = 3 ÷ 22 = **13.6%**, beyond the ~10% where grip peaks on many surfaces, so traction control cuts torque.

Example 3: control speed. A current loop running at 10 kHz updates every **100 µs**; a motor can go from zero to full torque in a few milliseconds, which is why EV traction control is so precise.

8. Real-world EV examples

- **One-pedal driving:** Nissan e-Pedal, Tesla's standard regen, Hyundai i-Pedal; Indian cars such as the Nexon EV and BE 6 offer selectable regen levels (often via paddles).
- **Torque vectoring:** Tesla Model S Plaid and Rivian quad-motor use separate rear motors per wheel.
- **Electric scooters:** many offer regen modes; benefit is smaller because of low mass and speed.

9. Manufacturing applications

- **Software calibration:** pedal maps, regen levels and limits are calibrated per variant; end-of-line flashing loads the correct software and parameters.

- **EOL checks:** resolver offset calibration (rotor position accuracy affects torque accuracy and efficiency), motor phase sequence check, roller-dyno drive and regen test.
- **Traceability:** software versions per VIN for OTA updates and recalls.

10. Common mistakes and misconceptions

- **"Regen means you rarely need brakes."** Friction brakes must still handle emergency stops, full batteries and low speed.
- **"More regen is always more efficient."** Coasting can be better than regenerating and then re-accelerating, because each conversion loses energy.
- **"The motor controller decides how fast the car goes."** The VCU decides; the MCU executes.
- **"Regen works the same when the battery is full or cold."** Charge acceptance is limited, so regen is reduced.

11. Practical case study: inconsistent brake feel

Problem. Test drivers report the brake pedal feeling different on cold mornings.

Root cause. With a cold battery, regen was limited; the blending software filled the gap with friction braking but with a short delay, so deceleration dipped briefly.

Fix. Predictive blending using BMS charge limits before braking starts, faster hydraulic response from the electric booster, and a smoother handover calibration.

Lesson. Brake blending is a vehicle-level integration problem across VCU, BMS and brake controller.

12. Revision notes

- VCU decides, MCU executes, BMS/ESC/thermal protect.
- FOC controls i_q (torque) and i_d (field) like a DC motor.
- Regen is limited by motor power, battery acceptance, stability and low speed.
- Brake blending gives consistent pedal feel.
- EV traction control acts in milliseconds.

13. Glossary

Term	Meaning
VCU	Vehicle control unit
MCU	Motor control unit
FOC	Field-oriented control
Torque arbitration	Choosing final torque from competing requests and limits
Brake blending	Sharing braking between regen and friction
One-pedal driving	Strong lift-off regen allowing most stops without the brake pedal
Torque vectoring	Different torque to left and right wheels

14. Knowledge check

Conceptual

1. Why does the VCU take the minimum of several torque limits?
2. What does FOC make easier?
3. Name four things that limit regen.
4. Why can EV traction control be finer than a petrol car's?
5. When is coasting better than regen?

Numerical

1. Regen power at 0.15 g from 60 km/h for 1,500 kg?
2. If the battery accepts only 25 kW, what share goes to friction brakes?
3. Wheel speed 18 m/s, vehicle 16 m/s. Slip ratio?

Application

1. Design three regen levels for an Indian city car and justify each.
2. Which data would you log to investigate a "car jerks when braking" complaint?

15. Answer key

Conceptual

1. Every limit protects something (battery, motor, inverter, stability); the smallest one is the safe torque.
2. It turns AC motor control into two independent DC-like loops for torque and field, giving fast, accurate torque.
3. Battery charge acceptance (SOC, temperature), motor/inverter power and torque limits, low vehicle speed, stability (slippery roads, rear-axle regen), driver regen setting.
4. Motor torque can be changed in milliseconds directly at the wheels, without throttle, combustion or gearbox delays.
5. When you will need the speed again soon (e.g. approaching a gentle slope or traffic that will move): coasting avoids two conversion losses.

Numerical

1. F = 1,500 × 9.81 × 0.15 = 2,207 N; P = 2,207 × 16.67 = **36.8 kW**.
2. 36.8 − 25 = 11.8 kW → **about 32%** to friction.
3. (18 − 16) ÷ 16 = **12.5%**.

Application

1. Low (~0.05 g): coasting feel for highways. Medium (~0.1 g): normal city driving, similar to engine braking. High/one-pedal (~0.2 g): stop-go traffic, maximum recovery. Adjust for battery temperature and SOC limits.
2. Pedal positions, VCU torque request and arbitration limits, MCU torque achieved, BMS charge limit, brake pressure, wheel speeds, ESC flags, vehicle deceleration (IMU), battery temperature and SOC, software versions.

16. Further learning

- Krishnan, *Electric Motor Drives: Modeling, Analysis and Control*.
- Texas Instruments and STMicroelectronics FOC application notes (free).
- Bosch Automotive Handbook: brake systems and ESC.
- SAE papers on regenerative brake blending.

Next: Module 6, the HV and LV electrical system.